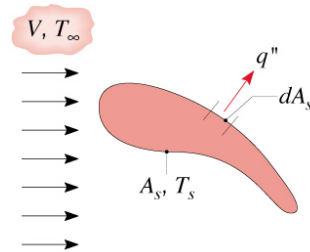


14. External flow correlations

Problem: Heat/momentum transfer by external flow



[Main idea]

Dimensional analysis shows

$$C_f, C_D = f(\text{Re, dimensionless location } x^*, \theta, \text{etc.})$$

$$\bar{C}_f, \bar{C}_D = f(\text{Re})$$

$$Nu = f(\text{Re, Pr, dimensionless location } x^*, \theta, \text{etc.})$$

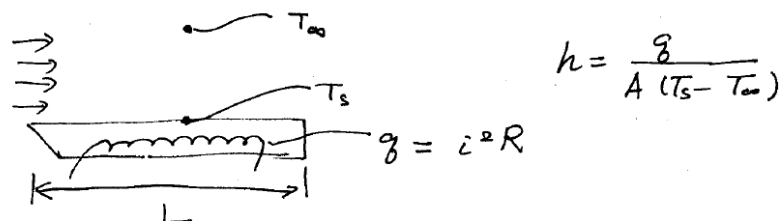
$$\bar{Nu} = f(\text{Re, Pr})$$

Note

- Friction coefficient: $C_f \equiv \frac{\tau}{\rho u_\infty^2 / 2}$
- Drag coefficient: $C_D \equiv \frac{(F_D / A_f)}{\rho u_\infty^2 / 2}$ ← Drag is by pressure (normal stress) as well as by shear (tangential stress)

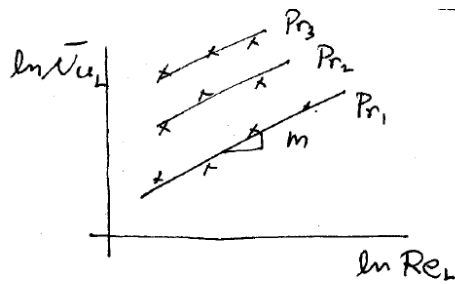
[Empirical method]

(Example)



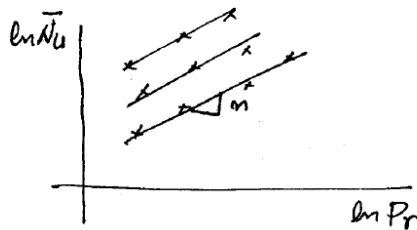
$$\bar{Nu}_L = c \text{Re}^m \text{Pr}^n \quad (A)$$

In the above equation, c , m and n are obtained by experiment



$$\ln \bar{Nu}_L = m \ln Re_L + \ln(c Pr^n)$$

↑
slope



$$\ln \bar{Nu}_L = n \ln Pr + \ln(c Re_L^m)$$

↑
slope

- Once equation (a) is obtained, it can be used for heat transfer calculation for any problem with the same geometry (shape, orientation, NOT size)
- Similar correlations can be obtained for various problems, e.g., for shear stress prediction.
- Property evaluation should be done at a well-defined temperature. Note that, because $T_s \neq T_\infty$, Re and Pr vary depending on which temperature you use for evaluating the properties. If a correlation does not specify the reference temperature use the “film temperature”.

Def) Film temperature $T_f \equiv \frac{T_s + T_\infty}{2}$

- HT and fluid mechanics correlations vary substantially depending on the flow mode (laminar vs. turbulent flow) → Need to calculate Re first to find the flow mode

[General methodology for computation]

1. Find geometry (plate, sphere, cylinder,?)
Identify BCs (constant Temp, heat flux, ...)
2. Determine the reference temperature (often “film temperature $T_f \equiv \frac{T_s + T_\infty}{2}$ ”)
Evaluate the properties at the reference temperature
→ Re , Pr
3. Determine the flow mode from Re (laminar, turbulent?)
4. Find a proper correlation for $C_f, C_D, \bar{C}_f, \bar{C}_D, Nu, \bar{Nu}_L, \dots$ (**Table 7.7**)
5. Compute τ, q, h , etc.
6. Check the validity of the correlations used in the analysis

Depending on the problem other dimensionless parameters may be important:

- Turbulence: ε/L
- Compressible flow: Ma
- Flow with a free surface: $Fr = V/\sqrt{gL}$
- ...